Table 1: S. cerevisiae datasets (pre-build, 2026-10-15; S. cerevisiae) built as LMDBs in the TorchCell dev tree. *Genotypes* and *Env* count distinct stored genotype and environment payloads; *Instances* is the record count; *Signal (gzip, bits)* the gzip size in bits of the concatenated stored instances (perturbation + environment + phenotype), as in the yeast table.

Dataset	Host	Genotypes	Env	Instances	Phenotype	Shape	Graph role	Signal (gzip, bits)
Environmental / chemogenomic response								
CellYieldDeChiara2022	*S. cerevisiae*	906	43	36,286	environment_response (cell_yield)	scalar	global	1.7×10^{7}
CrisprMagicLian2019	*S. cerevisiae*	266,304	3	266,304	environment_response (log2_ratio)	scalar	global	1.3×10^{8}
CrispriChemgenSmith2016	*S. cerevisiae*	1,035	26	14,463	environment_response (log2_ratio)	scalar	global	7.5×10^{6}
CrispriMormino2022	*S. cerevisiae*	12	1	12	environment_response (categorical)	scalar	global	1.4×10^{4}
DoublingTimeDeChiara2022	*S. cerevisiae*	906	43	36,295	environment_response (growth_rate)	scalar	global	1.7×10^{7}
EnvChemgenAuesukaree2009	*S. cerevisiae*	333	6	525	environment_response (categorical)	scalar	global	3.1×10^{5}
EnvChemgenCostanzo2021	*S. cerevisiae*	4,422	14	61,640	environment_response (differential_fitness)	scalar	global	1.1×10^{8}
EnvChemgenHoepfner2014	*S. cerevisiae*	10,839	5,586	30,308,574	environment_response (sensitivity_score)	scalar	global	2.1×10^{10}
EnvChemgenMota2024	*S. cerevisiae*	607	3	1,289	environment_response (ordinal)	scalar	global	7.4×10^{5}
EnvChemgenParsons2006	*S. cerevisiae*	4,111	82	329,254	environment_response (log2_ratio)	scalar	global	2.2×10^{8}
EnvChemgenPiotrowski2017	*S. cerevisiae*	289	11,949	3,435,820	environment_response (z_score)	scalar	global	1.4×10^{9}
EnvChemgenVanacloig2022	*S. cerevisiae*	3,649	45	164,205	environment_response (log2_ratio)	scalar	global	9.3×10^{7}
EnvChemgenWildenhain2015	*S. cerevisiae*	258	5,178	449,033	environment_response (z_score)	scalar	global	2.6×10^{8}
EnvChemgenYoshikawa2009	*S. cerevisiae*	4,729	4	37,832	environment_response (growth_rate)	scalar	global	1.0×10^{7}
FattyAcidSmith2006	*S. cerevisiae*	4,770	3	14,310	environment_response (ordinal)	scalar	global	1.8×10^{6}
HetHillenmeyer2008	*S. cerevisiae*	17,659	512	3,074,132	environment_response (log2_ratio)	scalar	global	2.0×10^{9}
HomHillenmeyer2008	*S. cerevisiae*	9,220	276	1,228,676	environment_response (z_score)	scalar	global	8.4×10^{8}
IsolateGrowthPeter2018	*S. cerevisiae*	971	35	33,985	environment_response (condition_growth_ratio)	scalar	global	1.6×10^{7}
Expression (RNA-seq)								
Boocock2025	*S. cerevisiae*	79,516	2	79,516	umi_count (scrnaseq_umi_count)	vector (3797)	node	8.9×10^{9}
CaudalPanTranscriptome2024	*S. cerevisiae*	943	1	943	expression_tpm (rnaseq_tpm)	vector (6017)	node	1.7×10^{9}
NadalRibellesPerturbSeq2025	*S. cerevisiae*	3,150	2	6,188	expression_log2_ratio (pseudobulk_scrnaseq_log2fc)	vector (5818)	node	2.5×10^{9}
Nguessan2025	*S. cerevisiae*	2,951	1	2,951	umi_count (scrnaseq_umi_count)	vector (1670)	node	4.5×10^{8}
PooledPromoterExpressionVaishnav2022	*S. cerevisiae*	71,158	1	71,158	expression_level	scalar	node	1.3×10^{8}
TfDeletionPseudobulkJackson2020	*S. cerevisiae*	72	11	784	umi_count (scrnaseq_umi_count)	vector (5075)	node	9.7×10^{7}
Expression (microarray)								
DmMicroarraySameith2015	*S. cerevisiae*	72	1	72	expression_log2_ratio	vector (6169)	node	1.7×10^{8}
InductionTimecourseHackett2020	*S. cerevisiae*	214	31	1,752	log2_cleaned	vector (6175)	node	4.7×10^{9}
MicroarrayKemmeren2014	*S. cerevisiae*	1,484	1	1,484	expression_log2_ratio	vector (6169)	node	3.2×10^{9}
MicroarrayScoreHu2007	*S. cerevisiae*	263	1	263	expression_score (error_model_x_score_weighted_duplicate_average)	vector (6371)	node	2.5×10^{8}
SmMicroarraySameith2015	*S. cerevisiae*	82	1	82	expression_log2_ratio	vector (6169)	node	1.8×10^{8}
Fitness								
Bloom2019	*S. cerevisiae*	13,950	38	530,100	environment_response (control_regression_residual)	scalar	global	2.6×10^{8}
CrispriSegregantHale2024	*S. cerevisiae*	895,494	2	8,692,635	environment_response (relative_fitness)	scalar	global	6.7×10^{9}
DmfCostanzo2016	*S. cerevisiae*	20,705,612	2	20,705,612	fitness	scalar	global	7.1×10^{9}
DmfKuzmin2018	*S. cerevisiae*	410,571	1	410,571	fitness	scalar	global	1.5×10^{8}
DmfKuzmin2020	*S. cerevisiae*	608,805	1	632,998	fitness	scalar	global	2.4×10^{8}
SmfBaryshnikova2010	*S. cerevisiae*	6,023	2	6,023	fitness	scalar	global	1.8×10^{6}
SmfCostanzo2016	*S. cerevisiae*	10,277	2	20,484	fitness	scalar	global	5.4×10^{6}
SmfKuzmin2018	*S. cerevisiae*	1,539	1	1,539	fitness	scalar	global	3.9×10^{5}
SmfKuzmin2020	*S. cerevisiae*	472	1	472	fitness	scalar	global	9.3×10^{4}
SmfODuibhir2014	*S. cerevisiae*	1,312	1	1,312	fitness	scalar	global	2.0×10^{6}
TmfKuzmin2018	*S. cerevisiae*	91,111	1	91,111	fitness	scalar	global	3.9×10^{7}
TmfKuzmin2020	*S. cerevisiae*	290,355	1	301,798	fitness	scalar	global	1.3×10^{8}
Gene interaction								
DmiCostanzo2016	*S. cerevisiae*	20,705,612	2	20,705,612	gene_interaction	scalar	edge	6.0×10^{9}
DmiKuzmin2018	*S. cerevisiae*	410,399	1	410,399	gene_interaction	scalar	edge	1.3×10^{8}
DmiKuzmin2020	*S. cerevisiae*	608,604	1	632,797	gene_interaction	scalar	edge	2.1×10^{8}
TmiKuzmin2018	*S. cerevisiae*	91,111	1	91,111	gene_interaction	scalar	hyperedge	3.4×10^{7}
TmiKuzmin2020	*S. cerevisiae*	290,355	1	301,798	gene_interaction	scalar	hyperedge	1.1×10^{8}
Genome architecture								
GenomeArchitecturePuddu2019	*S. cerevisiae*	4,550	1	8,835	genome_architecture	vector (28)	global	1.2×10^{7}
Life cycle								
LifeCycleDeChiara2022	*S. cerevisiae*	905	5	4,039	life_cycle	vector (4)	global	1.3×10^{6}
Metabolite								
AminoAcidCooper2010	*S. cerevisiae*	4,334	1	4,382	metabolite_level (ce_lif_peak_area_ratio_to_plate_mean)	vector (16)	bipartite node	5.2×10^{6}
AminoAcidMulleder2016	*S. cerevisiae*	4,678	1	4,678	metabolite_level (intracellular_concentration_mM)	vector (19)	bipartite node	1.1×10^{7}
BetaxanthinCachera2023	*S. cerevisiae*	4,759	1	4,759	metabolite_level (cri_spa_corrected_hsv_yellowness_24h)	scalar	bipartite node	4.5×10^{6}
FattyAcidXue2025	*S. cerevisiae*	176	1	176	metabolite_level (titer_mg_per_l)	vector (5)	bipartite node	1.9×10^{5}
IsobutanolScreenLopez2024	*S. cerevisiae*	4,607	1	4,805	metabolite_level (biosensor_gfp_fluorescence_fold_change)	scalar	bipartite node	9.0×10^{5}
IsobutanolValidatedLopez2024	*S. cerevisiae*	226	1	227	metabolite_level (biosensor_gfp_fluorescence_fold_change)	scalar	bipartite node	7.4×10^{4}
MetaboliteDaSilveira2014	*S. cerevisiae*	127	1	127	metabolite_level (lipidomics_ms_relative_abundance_au)	vector (135)	bipartite node	1.1×10^{6}
MetaboliteZelezniak2018	*S. cerevisiae*	95	1	129	metabolite_level (lc_srm_signal_batch_corrected_uncalibrated_dataset1)	vector (14)	bipartite node	3.5×10^{5}
OrganicAcidYoshida2012	*S. cerevisiae*	17	1	17	metabolite_level (hplc_organic_acid_titer_mM)	vector (6)	bipartite node	3.3×10^{4}
Morphology								
CarotenoidOzaydin2013	*S. cerevisiae*	4,475	1	4,607	visual_score	scalar	global	7.7×10^{6}
ScmdOhnuki2018	*S. cerevisiae*	1,112	1	1,112	calmorph	vector (281)	global	4.9×10^{7}
ScmdOhnuki2022	*S. cerevisiae*	1,982	1	1,982	calmorph	vector (281)	global	8.8×10^{7}
ScmdOhya2005	*S. cerevisiae*	4,718	1	4,718	calmorph	vector (281)	global	1.5×10^{8}
Protein abundance								
ProteomeMessner2021	*S. cerevisiae*	1	18	70	protein_abundance (swath_ms_diann_maxlfq_proteotypic_quantity)	vector (1920)	node	1.2×10^{7}
ProteomeMessner2023	*S. cerevisiae*	4,549	1	4,699	protein_abundance (swath_ms_maxlfq_batch_corrected_quantity)	vector (1830)	node	1.2×10^{9}
ProteomeMuenzner2024	*S. cerevisiae*	795	1	795	protein_abundance (swath_ms_maxlfq_precursor_batch_corrected_quantity)	vector (1542)	node	9.6×10^{8}
ProteomeZelezniak2018	*S. cerevisiae*	97	1	97	protein_abundance (swath_ms_label_free_log_signal_sva)	vector (726)	node	1.7×10^{7}
Protein synthesis / turnover								
ProteinTurnoverMuenzner2024	*S. cerevisiae*	55	1	55	degradation_rate (dynamic_silac_lys8_kdp_per_hour_no_dilution_correction)	vector (1015)	node	6.9×10^{7}
Viability								
GeneEssentialitySgd	*S. cerevisiae*	1,180	5	1,329	is_essential	scalar	node	6.8×10^{5}
SynthLethalityYeastSynthLethDb	*S. cerevisiae*	14,000	1	14,000	is_synthetic_lethal	scalar	edge	8.0×10^{6}
SynthRescueYeastSynthLethDb	*S. cerevisiae*	6,948	1	6,948	is_synthetic_rescue	scalar	edge	4.0×10^{6}
Total (69 datasets, 69 probed)				**93,266,786**				7.2×10^{10}

Table 2: Signal controls, same pass. *Structure*: leaves replaced by type tokens. *Shuffled*: numeric leaves permuted within key path. *Noise*: digits of numeric leaves redrawn uniformly at stored precision. *Binary*: per-path z-score thresholded at 0, written in place. *Packed 1-bit*: gzip of those bits alone, 8 per byte.

Dataset	Instances	Numeric leaves	Signal (gzip, bits)	Structure (bits)	Shuffled (bits)	Noise (bits)	Binary (bits)	Packed 1-bit (bits)	Structure/Signal	Signal/Noise	Packed/Signal
Environmental / chemogenomic response											
CellYieldDeChiara2022	36,286	72,572	1.7×10^{7}	1.2×10^{6}	1.7×10^{7}	1.7×10^{7}	1.4×10^{7}	3.5×10^{4}	0.07	0.99	0.00
CrisprMagicLian2019	266,304	1,331,520	1.3×10^{8}	2.2×10^{7}	1.3×10^{8}	1.4×10^{8}	5.8×10^{7}	7.4×10^{5}	0.16	0.93	0.01
CrispriChemgenSmith2016	14,463	72,315	7.5×10^{6}	7.3×10^{5}	7.6×10^{6}	8.2×10^{6}	5.1×10^{6}	2.2×10^{4}	0.10	0.92	0.00
CrispriMormino2022	12	24	1.4×10^{4}	5.9×10^{3}	1.4×10^{4}	1.4×10^{4}	1.4×10^{4}	1.8×10^{2}	0.43	0.98	0.01
DoublingTimeDeChiara2022	36,295	72,590	1.7×10^{7}	1.2×10^{6}	1.7×10^{7}	1.7×10^{7}	1.4×10^{7}	3.6×10^{4}	0.07	0.98	0.00
EnvChemgenAuesukaree2009	525	5,419	3.1×10^{5}	1.3×10^{5}	3.4×10^{5}	4.9×10^{5}	2.9×10^{5}	1.4×10^{3}	0.41	0.63	0.00
EnvChemgenCostanzo2021	61,640	1,466,135	1.1×10^{8}	3.3×10^{7}	1.3×10^{8}	1.7×10^{8}	9.9×10^{7}	2.6×10^{5}	0.31	0.64	0.00
EnvChemgenHoepfner2014	30,308,574	60,617,148	2.1×10^{10}	2.4×10^{9}	2.1×10^{10}	2.1×10^{10}	1.8×10^{10}	3.5×10^{7}	0.12	0.99	0.00
EnvChemgenMota2024	1,289	16,757	7.4×10^{5}	3.6×10^{5}	8.2×10^{5}	1.3×10^{6}	7.1×10^{5}	2.2×10^{3}	0.48	0.55	0.00
EnvChemgenParsons2006	329,254	987,762	2.2×10^{8}	2.6×10^{7}	2.2×10^{8}	2.4×10^{8}	1.9×10^{8}	6.9×10^{5}	0.12	0.94	0.00
EnvChemgenPiotrowski2017	3,435,820	6,871,640	1.4×10^{9}	2.3×10^{8}	1.4×10^{9}	1.5×10^{9}	1.3×10^{9}	3.9×10^{6}	0.16	0.96	0.00
EnvChemgenVanacloig2022	164,205	656,820	9.3×10^{7}	8.6×10^{6}	9.3×10^{7}	9.5×10^{7}	4.5×10^{7}	3.6×10^{5}	0.09	0.98	0.00
EnvChemgenWildenhain2015	449,033	898,066	2.6×10^{8}	3.7×10^{7}	2.6×10^{8}	2.7×10^{8}	2.4×10^{8}	3.6×10^{5}	0.14	0.97	0.00
EnvChemgenYoshikawa2009	37,832	75,664	1.0×10^{7}	3.0×10^{6}	1.0×10^{7}	1.1×10^{7}	9.7×10^{6}	1.3×10^{4}	0.30	0.90	0.00
FattyAcidSmith2006	14,310	28,620	1.8×10^{6}	6.7×10^{5}	1.8×10^{6}	2.3×10^{6}	1.7×10^{6}	8.5×10^{3}	0.38	0.76	0.00
HetHillenmeyer2008	3,074,132	7,586,524	2.0×10^{9}	2.7×10^{8}	2.0×10^{9}	2.0×10^{9}	1.7×10^{9}	6.3×10^{6}	0.14	0.97	0.00
HomHillenmeyer2008	1,228,676	3,266,922	8.4×10^{8}	1.1×10^{8}	8.4×10^{8}	8.6×10^{8}	7.0×10^{8}	2.9×10^{6}	0.13	0.97	0.00
IsolateGrowthPeter2018	33,985	67,970	1.6×10^{7}	1.3×10^{6}	1.6×10^{7}	1.6×10^{7}	1.4×10^{7}	3.5×10^{4}	0.08	0.96	0.00
Expression (RNA-seq)											
Boocock2025	79,516	360,790,503	8.9×10^{9}	2.4×10^{9}	9.4×10^{9}	1.1×10^{10}	3.8×10^{9}	2.3×10^{8}	0.27	0.82	0.03
CaudalPanTranscriptome2024	943	17,715,967	1.7×10^{9}	3.7×10^{8}	1.8×10^{9}	2.1×10^{9}	1.2×10^{9}	1.3×10^{7}	0.21	0.84	0.01
NadalRibellesPerturbSeq2025	6,188	36,405,529	2.5×10^{9}	8.2×10^{8}	2.5×10^{9}	2.6×10^{9}	9.4×10^{8}	3.1×10^{7}	0.33	0.95	0.01
Nguessan2025	2,951	16,459,410	4.5×10^{8}	1.9×10^{8}	4.7×10^{8}	5.1×10^{8}	2.5×10^{8}	1.1×10^{7}	0.42	0.88	0.02
PooledPromoterExpressionVaishnav2022	71,158	2,638,069	1.3×10^{8}	3.8×10^{7}	2.0×10^{8}	2.4×10^{8}	1.2×10^{8}	1.4×10^{5}	0.30	0.52	0.00
TfDeletionPseudobulkJackson2020	784	3,029,208	9.7×10^{7}	7.0×10^{7}	9.8×10^{7}	9.9×10^{7}	8.0×10^{7}	1.9×10^{6}	0.72	0.98	0.02
Expression (microarray)											
DmMicroarraySameith2015	72	2,220,912	1.7×10^{8}	4.9×10^{7}	1.7×10^{8}	1.7×10^{8}	5.2×10^{7}	1.6×10^{6}	0.29	0.98	0.01
InductionTimecourseHackett2020	1,752	84,572,800	4.7×10^{9}	1.8×10^{9}	4.8×10^{9}	5.2×10^{9}	1.9×10^{9}	5.1×10^{7}	0.39	0.90	0.01
MicroarrayKemmeren2014	1,484	45,775,464	3.2×10^{9}	1.0×10^{9}	3.3×10^{9}	3.3×10^{9}	1.1×10^{9}	2.7×10^{7}	0.32	0.98	0.01
MicroarrayScoreHu2007	263	3,344,872	2.5×10^{8}	7.5×10^{7}	2.5×10^{8}	2.6×10^{8}	8.1×10^{7}	3.3×10^{6}	0.30	0.97	0.01
SmMicroarraySameith2015	82	2,529,372	1.8×10^{8}	5.6×10^{7}	1.8×10^{8}	1.8×10^{8}	5.8×10^{7}	1.5×10^{6}	0.32	0.98	0.01
Fitness											
Bloom2019	530,100	1,060,200	2.6×10^{8}	2.0×10^{7}	2.6×10^{8}	2.7×10^{8}	2.3×10^{8}	5.5×10^{5}	0.08	0.97	0.00
CrispriSegregantHale2024	8,692,635	34,770,540	6.7×10^{9}	3.2×10^{8}	7.3×10^{9}	7.6×10^{9}	4.9×10^{9}	1.5×10^{7}	0.05	0.88	0.00
DmfCostanzo2016	20,705,612	106,311,976	7.1×10^{9}	9.4×10^{8}	7.3×10^{9}	8.7×10^{9}	4.5×10^{9}	5.5×10^{7}	0.13	0.82	0.01
DmfKuzmin2018	410,571	2,052,827	1.5×10^{8}	2.0×10^{7}	1.5×10^{8}	1.8×10^{8}	1.0×10^{8}	1.2×10^{6}	0.14	0.85	0.01
DmfKuzmin2020	632,998	3,164,990	2.4×10^{8}	3.2×10^{7}	2.4×10^{8}	2.8×10^{8}	1.6×10^{8}	1.8×10^{6}	0.13	0.85	0.01
SmfBaryshnikova2010	6,023	24,868	1.8×10^{6}	2.5×10^{5}	1.8×10^{6}	2.2×10^{6}	1.1×10^{6}	2.0×10^{4}	0.14	0.81	0.01
SmfCostanzo2016	20,484	105,352	5.4×10^{6}	7.9×10^{5}	6.3×10^{6}	7.9×10^{6}	3.9×10^{6}	6.1×10^{4}	0.14	0.69	0.01
SmfKuzmin2018	1,539	2,895	3.9×10^{5}	5.8×10^{4}	3.9×10^{5}	4.3×10^{5}	3.2×10^{5}	3.1×10^{3}	0.15	0.90	0.01
SmfKuzmin2020	472	2,360	9.3×10^{4}	1.5×10^{4}	1.0×10^{5}	1.5×10^{5}	6.7×10^{4}	2.3×10^{3}	0.17	0.61	0.02
SmfODuibhir2014	1,312	47,232	2.0×10^{6}	6.2×10^{5}	3.3×10^{6}	4.0×10^{6}	1.8×10^{6}	3.2×10^{3}	0.31	0.50	0.00
TmfKuzmin2018	91,111	455,555	3.9×10^{7}	6.6×10^{6}	4.0×10^{7}	4.5×10^{7}	2.7×10^{7}	2.6×10^{5}	0.17	0.87	0.01
TmfKuzmin2020	301,798	1,508,990	1.3×10^{8}	2.2×10^{7}	1.3×10^{8}	1.5×10^{8}	8.9×10^{7}	8.1×10^{5}	0.17	0.87	0.01
Gene interaction											
DmiCostanzo2016	20,705,612	44,195,140	6.0×10^{9}	1.0×10^{9}	6.0×10^{9}	6.5×10^{9}	4.7×10^{9}	4.4×10^{7}	0.17	0.92	0.01
DmiKuzmin2018	410,399	820,798	1.3×10^{8}	2.2×10^{7}	1.3×10^{8}	1.4×10^{8}	1.0×10^{8}	8.1×10^{5}	0.16	0.94	0.01
DmiKuzmin2020	632,797	1,265,594	2.1×10^{8}	3.3×10^{7}	2.1×10^{8}	2.2×10^{8}	1.6×10^{8}	1.3×10^{6}	0.16	0.94	0.01
TmiKuzmin2018	91,111	182,222	3.4×10^{7}	6.5×10^{6}	3.4×10^{7}	3.6×10^{7}	2.7×10^{7}	1.8×10^{5}	0.19	0.94	0.01
TmiKuzmin2020	301,798	603,596	1.1×10^{8}	2.2×10^{7}	1.1×10^{8}	1.2×10^{8}	9.0×10^{7}	6.0×10^{5}	0.19	0.95	0.01
Genome architecture											
GenomeArchitecturePuddu2019	8,835	256,215	1.2×10^{7}	5.3×10^{5}	1.2×10^{7}	1.6×10^{7}	3.3×10^{6}	1.9×10^{5}	0.05	0.72	0.02
Life cycle											
LifeCycleDeChiara2022	4,039	16,382	1.3×10^{6}	1.4×10^{5}	1.4×10^{6}	1.6×10^{6}	8.5×10^{5}	1.5×10^{4}	0.10	0.83	0.01
Metabolite											
AminoAcidCooper2010	4,382	136,404	5.2×10^{6}	4.0×10^{5}	5.2×10^{6}	6.6×10^{6}	1.2×10^{6}	8.8×10^{4}	0.08	0.80	0.02
AminoAcidMulleder2016	4,678	210,510	1.1×10^{7}	1.4×10^{6}	1.2×10^{7}	1.5×10^{7}	3.8×10^{6}	1.4×10^{5}	0.12	0.74	0.01
BetaxanthinCachera2023	4,759	52,349	4.5×10^{6}	1.9×10^{6}	5.0×10^{6}	6.1×10^{6}	3.5×10^{6}	2.7×10^{4}	0.41	0.74	0.01
FattyAcidXue2025	176	2,816	1.9×10^{5}	2.8×10^{4}	1.9×10^{5}	2.3×10^{5}	5.9×10^{4}	2.1×10^{3}	0.15	0.84	0.01
IsobutanolScreenLopez2024	4,805	14,415	9.0×10^{5}	1.8×10^{5}	9.2×10^{5}	1.1×10^{6}	5.3×10^{5}	7.4×10^{3}	0.20	0.81	0.01
IsobutanolValidatedLopez2024	227	908	7.4×10^{4}	1.2×10^{4}	7.6×10^{4}	8.6×10^{4}	3.4×10^{4}	3.7×10^{2}	0.16	0.86	0.00
MetaboliteDaSilveira2014	127	36,419	1.1×10^{6}	9.9×10^{4}	1.2×10^{6}	1.6×10^{6}	3.2×10^{5}	1.9×10^{4}	0.09	0.73	0.02
MetaboliteZelezniak2018	129	5,166	3.5×10^{5}	2.7×10^{4}	3.6×10^{5}	3.9×10^{5}	8.3×10^{4}	4.6×10^{3}	0.08	0.91	0.01
OrganicAcidYoshida2012	17	391	3.3×10^{4}	9.6×10^{3}	3.4×10^{4}	4.5×10^{4}	2.7×10^{4}	5.6×10^{2}	0.29	0.75	0.02
Morphology											
CarotenoidOzaydin2013	4,607	179,673	7.7×10^{6}	2.9×10^{6}	1.3×10^{7}	1.5×10^{7}	7.7×10^{6}	1.1×10^{4}	0.38	0.50	0.00
ScmdOhnuki2018	1,112	564,896	4.9×10^{7}	8.9×10^{5}	5.0×10^{7}	5.4×10^{7}	6.4×10^{6}	5.4×10^{5}	0.02	0.92	0.01
ScmdOhnuki2022	1,982	994,962	8.8×10^{7}	1.4×10^{6}	8.9×10^{7}	9.4×10^{7}	9.2×10^{6}	9.3×10^{5}	0.02	0.93	0.01
ScmdOhya2005	4,718	2,363,718	1.5×10^{8}	2.9×10^{6}	1.5×10^{8}	1.7×10^{8}	2.0×10^{7}	2.3×10^{6}	0.02	0.91	0.02
Protein abundance											
ProteomeMessner2021	70	277,682	1.2×10^{7}	1.1×10^{6}	1.2×10^{7}	1.3×10^{7}	4.9×10^{6}	1.4×10^{5}	0.09	0.97	0.01
ProteomeMessner2023	4,699	16,955,915	1.2×10^{9}	5.2×10^{7}	1.2×10^{9}	1.2×10^{9}	3.5×10^{8}	8.7×10^{6}	0.05	0.96	0.01
ProteomeMuenzner2024	795	7,685,904	9.6×10^{8}	1.1×10^{8}	9.8×10^{8}	1.2×10^{9}	8.2×10^{8}	3.4×10^{6}	0.12	0.77	0.00
ProteomeZelezniak2018	97	211,363	1.7×10^{7}	2.8×10^{5}	1.8×10^{7}	1.7×10^{7}	3.3×10^{6}	1.4×10^{5}	0.02	1.00	0.01
Protein synthesis / turnover											
ProteinTurnoverMuenzner2024	55	539,502	6.9×10^{7}	7.2×10^{6}	7.1×10^{7}	8.9×10^{7}	5.7×10^{7}	2.5×10^{5}	0.10	0.78	0.00
Viability											
GeneEssentialitySgd	1,329	3	6.8×10^{5}	2.6×10^{5}	6.8×10^{5}	6.8×10^{5}	6.8×10^{5}	1.7×10^{2}	0.38	1.00	0.00
SynthLethalityYeastSynthLethDb	14,000	14,000	8.0×10^{6}	2.3×10^{6}	8.1×10^{6}	8.3×10^{6}	7.9×10^{6}	1.0×10^{4}	0.29	0.96	0.00
SynthRescueYeastSynthLethDb	6,948	0	4.0×10^{6}	1.2×10^{6}	4.0×10^{6}	4.0×10^{6}	4.0×10^{6}	1.6×10^{2}	0.29	1.00	0.00
Total (69 probed)	**93,266,786**	**886,671,302**	7.2×10^{10}	1.3×10^{10}	7.4×10^{10}	7.9×10^{10}	4.8×10^{10}	5.6×10^{8}	**0.18**	**0.91**	**0.01**